time machine. A device for traveling backward in time. In physicists' jargon, a "closed timelike curve."

topology. The branch of mathematics that deals with the qualitative ways that objects are connected to each other or to themselves. For example, topology distinguishes a sphere (which has no hole) from a doughnut (which has one).

tritium. Atomic nuclei made of one proton and two neutrons bound together by the nuclear force.

ultraviolet radiation. Electromagnetic radiation with a wavelength a little shorter than light; see Figure P.2 on page 25.

uncertainty principle. A quantum mechanical law which states that, if one measures the position of an object or the strength of a field with high precision, one's measurement must necessarily perturb the object's velocity or the field's rate of change by an unpredictable amount.

universe. A region of space that is disconnected from all other regions of space, much as an island is disconnected from all other pieces of land.

Universe. Our universe.

unstable. The property of an object that if one perturbs it slightly, the perturbation will grow large, thereby changing the object greatly and perhaps even destroying it. Also called, in more complete terminology, "unstable against small perturbations."

uranium-255. A specific type of uranium nucleus which contains 235 protons and neutrons (92 protons and 143 neutrons).

vacuum. A region of spacetime from which have been removed all the particles and fields and energy that one can remove; the only things left are the irremovable vacuum fluctuations.

vacuum fluctuations. Random, unpredictable, irremovable oscillations of a field (for example, an electromagnetic or gravitational field), which are caused by a tug-of-war in which small regions of space momentarily steal energy from adjacent regions and then give it back. See also *vacuum* and *virtual particles.*

virtual particles. Particles that are created in pairs using energy borrowed from a nearby region of space. The laws of quantum mechanics require that the energy be given back quickly, so the virtual particles annihilate quickly and cannot be captured. Virtual particles are the particle aspect of vacuum fluctuations, as seen by freely falling